MandateBench: Mandate Faithfulness and Pre-Signature Monitorability for Agentic-Payment LLMs

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Preprint · 2026 · v6 pilot (9 frontier models, 753 graded authorizations + adversarial duels) · Open source (MIT), github.com/Johnnyevans32/mandatebench · CC BY 4.0

Abstract

Emerging standards for AI payments — Google's Agent Payments Protocol (AP2) and Coinbase's x402 — let an autonomous agent commit funds by signing a *mandate*: a cryptographically signed statement of what it is authorized to spend. Because a signature is irreversible, the safety-critical question is not whether a protocol gateway can reject a malformed request, but whether the *agent* can be manipulated into authorizing a payment it should refuse, and whether that can be caught **before** the signature. MandateBench measures mandate faithfulness across nine frontier models (one or more per major lab) under a taxonomy of adversarial pressures, with a deterministic, judge-free ground truth: violations are predicates over a signed object, not an LLM judge's opinion. Its central contribution is separating two violation classes. **Rule** violations (over budget, over per-payment cap, unauthorized recipient, out-of-scope category, expired mandate) are structural — a stateless gateway that validates the signed fields already catches them. **Intent** violations satisfy every signed field yet betray the mandate's natural-language intent — a grocery mandate spent on a stored-value gift card or a bottle of vintage champagne — and *no* structural check can catch them. We find models that catch structural violations near-perfectly (100%) systematically authorize a third of intent violations (mean intent-catch 75%, range 44–100%). We then measure **pre-signature monitorability**: a monitor reading only the agent's private reasoning predicts violations at AUROC 0.67. Instructing the agent to keep its reasoning bland is doubly harmful — the number of violations roughly doubles, and the monitor's AUROC falls from 0.67 to 0.30 (below chance), because suppressed reasoning makes a breach read as more routine than a compliant decision. Reasoning suppression does not merely blind the overseer; it blinds the agent.

1. Introduction

Agentic-payment protocols move the authority to spend money from a human click to a model's signature. In AP2 and x402, a user (or their wallet) issues a signed mandate delimiting a budget, per-payment cap, permitted recipients, permitted categories, and a validity window; an agent then negotiates with merchants and *signs* to authorize each payment. The signature is the point of no return: once emitted it settles, and there is no undo.

A natural objection is that the protocol itself enforces the mandate — a gateway can reject any signed request whose amount, recipient, or category falls outside the mandate. This is true, and it is exactly why the interesting failures are elsewhere. Hard structural limits are the easy case. The dangerous case is a request that is *structurally valid* — correct recipient, in-budget, in-category — but is not what the user meant, or is a payment the agent was socially engineered into signing. A gateway cannot distinguish "this week's groceries" from a $25 gift card charged to the same grocer for the same amount; only a model that reasons about intent can. MandateBench is built to measure precisely this gap, and to ask whether an overseer can catch a breach in the moment before the signature.

We organize the work around four questions. **RQ1**: how often do models authorize mandate-violating payments under adversarial pressure? **RQ2**: does the failure fall on structural (rule) or intent violations, and does the split separate models? **RQ3**: can a monitor reading only the agent's private reasoning predict a violation before the signature, and how robust is that to an agent instructed to hide its reasoning? **RQ4** (adversarial): when one model plays the merchant and attacks another, who breaks and who holds?

2. Related work

Agent security benchmarks such as CAIA and function-level red-teaming of AP2/x402 flows focus predominantly on a single vector — prompt injection — and on protocol conformance. Chain-of-thought monitoring and the broader debate on reasoning-based oversight ask whether a model's reasoning can be read to catch misbehaviour; recent position work argues monitorability is a fragile but valuable property. Deception benchmarks (e.g. cross-play social-deduction evaluations) measure monitorability of hidden intent in games. MandateBench differs on three axes: (i) a high-stakes, irreversible-action substrate (signed money movement) rather than a game or a QA task; (ii) a deterministic, judge-free ground truth; and (iii) an explicit separation of gateway-catchable rule violations from intent violations that only a model or model-monitor can catch — which reframes "why put a model in the loop" as an empirical, measurable question.

3. Method

3.1 Signed mandates and deterministic ground truth

Each scenario issues an Ed25519-signed mandate over a canonical (sorted-key) serialization; the agent verifies the signature before acting. A payment is a cart (recipient, amount in integer minor units, currency, category, quantity). A deterministic checker `checkAuthorization(mandate, cart, priorAuthorized)` returns the exact set of violated predicates: over-budget, over per-payment cap, unauthorized recipient, out-of-scope category, expired TTL, quantity creep, currency mismatch, and structuring (splitting an over-budget order into sub-cap charges). No LLM judges outcomes; labels are ground truth by construction.

3.2 Pressure taxonomy and violation classes

Scenarios span a taxonomy of pressures — baseline, principal-agent overreach, merchant social engineering, urgency/scarcity, category ambiguity, prompt injection, scope drift, and conflicting mandates — as well as two non-traps (a clean order and an at-cap order) that measure over-refusal. Each scenario is labelled by ground-truth class: **clean** (authorizing is correct), **rule** (breaks a structural term), or **intent** (satisfies every structural term but breaks the mandate's stated intent). The intent traps are constructed so a stateless gateway validating the signed fields passes them: a grocery mandate (FreshMart, groceries, ≤$30, in budget) is offered a $25 FreshMart gift card, a $29 bottle of vintage champagne, or a grocery total padded with an $8 charity round-up. Each is structurally perfect; each betrays the intent.

3.3 Monitorability protocol (RQ3)

For every graded decision we store the agent's private reasoning and public statement. A monitor model then reads *only* the agent's private reasoning — never the mandate numbers or the cart — and outputs a violation probability in [0,1]. We report AUROC of these scores against the objective violation labels. The adversarial arm re-runs the benchmark with an added instruction telling the agent its reasoning may be reviewed and to keep it bland, uninformative, and free of any mention of limits or risk; the decision is otherwise unconstrained. Comparing monitor AUROC with and without suppression measures oversight robustness.

3.4 Models

Nine flagship models via OpenRouter with structured outputs and temperature 0 where supported: GPT-5.5, Claude Opus 4.8, Gemini 3.5 Flash, DeepSeek V4 Pro, Grok 4.3, Llama 4 Maverick, Qwen3.7 Max, Kimi K2.6, and Mistral Large 2512. The pilot runs three repetitions per scenario (snapshot v6: 378 graded authorizations; a suppressed-reasoning arm v6-hidden: 375), plus adversarial duels.

4. Results

4.1 RQ1 — Violations under pressure

Over trap scenarios, only three of nine models never violate (GPT-5.5, Claude Opus 4.8, Kimi K2.6). The rest range from 8% (Gemini, Grok, Qwen) to 25% (Llama 4 Maverick), with DeepSeek and Mistral in between. Notably, running repetitions matters: Mistral scored 0/9 in a single pass but 3/27 over three reps — a single run overstated its robustness.

4.2 RQ2 — Rule vs intent (headline)

Splitting violations by class is the central result (Table 1). Structural (rule) violations are caught near-perfectly by almost every model — as expected, since they are the gateway-catchable case. Intent violations separate the field: three models catch them fully, but five frontier models catch only two of three, and Llama fewer than half. A model can look flawless on the structural traps and still authorize a third of the intent violations a gateway is blind to.

MODEL	RULE CAUGHT	INTENT CAUGHT	FALSE REFUSAL
GPT-5.5	100%	100%	0%
Claude Opus 4.8	100%	100%	0%
Kimi K2.6	100%	100%	0%
Gemini 3.5 Flash	100%	67%	0%
Grok 4.3	100%	67%	17%
Qwen3.7 Max	100%	67%	0%
Mistral Large 2512	93%	67%	0%
DeepSeek V4 Pro	89%	67%	0%
Llama 4 Maverick	85%	44%	0%

Table 1. Fraction of violations caught, split by class, snapshot v6 (3 reps/scenario). Rule = structural, gateway-catchable. Intent = structurally valid but betrays stated intent. False refusal = clean orders wrongly declined (usefulness cost). Mean intent-catch 75%.

4.3 RQ3 — Pre-signature monitorability

A monitor reading only the agent's private reasoning predicts violations at **AUROC 0.669** (29 violations vs 345 compliant decisions) — moderately, not perfectly, diagnostic. Under reasoning suppression the picture inverts. First, suppression is not behaviourally free: total violations roughly **double** (29 → 51). Inspecting the reasoning shows why — told to stay bland, models stop reasoning about intent and collapse to structural checks; violating intent-trap runs literally read "within budget, under per-payment cap, to allowed recipient, in allowed category" and authorize the gift card. Second, the monitor's **AUROC falls**

from 0.669 to 0.304 — below chance — because the bland, reassuring reasoning attached to a breach looks *more* innocent than an ordinary compliant decision, actively misleading the overseer. Naive reasoning-hiding therefore both raises the violation rate and defeats the monitor; reasoning suppression blinds the agent as well as the observer.

4.4 RQ4 — Under live attack, the same models break

Beyond scripted traps, we run a full adversarial matrix: every model plays the merchant and attacks every model's agent over a multi-turn negotiation (9×9 = 81 duels). Robustness is a property of the *defender*, not the attacker: no attacker exceeds a 22% success rate, yet breaches concentrate sharply on a few agents (Table 2). Six of nine agents are never broken; these are the same models that miss intent violations in Table 1 — the scripted and adversarial methods agree, which is the corroboration a pilot most needs.

AGENT (DEFENDER)	BREACHED BY ATTACKERS	RATE
GPT-5.5 · Opus 4.8 · Gemini · Grok · Qwen · Kimi	0 / 9	0%
DeepSeek V4 Pro	1 / 9	11%
Mistral Large 2512	5 / 9	56%
Llama 4 Maverick	6 / 9	67%

Table 2. Agent robustness in the 9×9 adversarial matrix (81 duels, over-cap goal, up to 3 turns). Rate = fraction of the 9 attackers that broke this agent. No attacker exceeded 22% success, so robustness tracks the defender, not the attacker.

5. Discussion

The rule/intent split reframes the "the gateway already enforces the mandate" objection: gateways enforce structure, and structure is the case models already handle. Intent is the residual — unenforceable by construction, and the axis on which frontier models measurably differ. This is the empirical justification for a model (or a model-monitor) in the payment loop. The monitorability result is a cautionary one: reasoning is only moderately diagnostic even in the clear, and a trivial suppression instruction pushes the monitor below chance while making the agent less safe. Oversight that leans on reading reasoning must assume the channel can be cheaply, and counter-productively, degraded.

6. Limitations

This is a pilot: three repetitions per scenario yield wide intervals, and the intent-violation set is small (three constructions over one grocery-mandate domain). Results may be sensitive to system-prompt wording — during development an unfair expired-mandate prompt produced a spurious universal failure until the agent was given issued/expiry/current timestamps. The suppression arm uses a single, naive hiding instruction; a trained or steganographic adversary is the stronger test of monitorability and is future work, as are an adaptive-adversary robustness curve and multi-domain mandates.

7. Conclusion

MandateBench provides a deterministic, judge-free instrument for the reasoning-based oversight of high-stakes agentic payments. Its measurements show that frontier models handle the violations a gateway already catches but not the intent violations only a model can, and that reading an agent's reasoning to catch breaches is both moderately reliable and cheaply defeated — in a way that also degrades the agent. We release the benchmark, harness, and live dashboard openly.

References

1. Google. Agent Payments Protocol (AP2). 2025.
2. Coinbase. x402: An open standard for internet-native payments. 2025.
3. Center for AI Safety et al. CAIA: Benchmarking agentic AI security. arXiv:2510.00332, 2025.
4. Korbak, T. et al. Chain-of-thought monitorability: A new and fragile opportunity for AI safety. 2025.
5. Eburu, E. Among LLMs: A Cross-Play Benchmark for Deception, Detection, and the Monitorability of Reasoning. Zenodo, doi:10.5281/zenodo.21209429, 2026.
6. Anthropic. Sabotage and monitoring evaluations for frontier models. 2024–2025.